

Independence, Basis and Dimension

Subspaces and their true dimension

* A matrix A of size $m \times n$ has n columns, but the column space $C(A)$ - the subspace spanned by these columns - doesn't always have a dimension of n .

Hence, the true dimension of the column space is determined by the number of linearly independent columns, not just (the total number of columns) *

Suppose A is a matrix m by n with $m < n$. Then there are non zero solutions to $Ax = 0$.
(basically, more unknowns than equations)

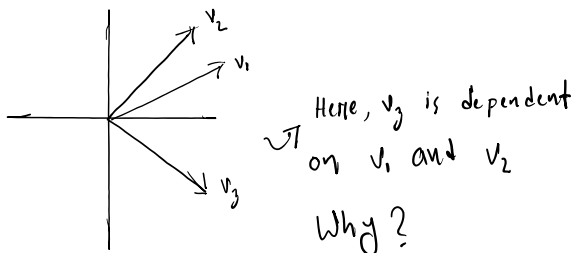
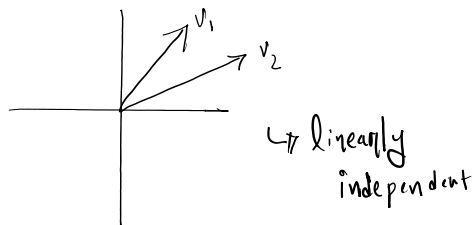
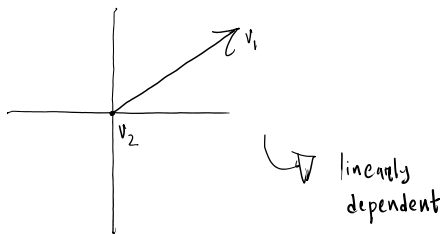
Since, there are n variables and at most m pivots, so there would be at least $n - m$ free variables.

* Every vector in the space is a unique combination of the basis vectors.

↳ Independence

Vectors $x_1, x_2, \dots, x_n$ are independent if no combination gives zero vector. (Except the zero combination)

$$c_1 x_1 + c_2 x_2 + \dots + c_n x_n \neq 0 \quad (\text{where all } c_i \neq 0)$$



Three vectors on a plane have to be dependent on each other.

$$A = \begin{bmatrix} v_1 & v_2 & v_3 \\ 2 & 1 & 5/2 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \hookrightarrow \text{In } \mathbb{R}^2, \text{ any three vectors } (a, b), (c, d) \text{ and } (e, f) \text{ are linearly dependent}$$

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$\hookrightarrow$ There are some combination that gives the zero vector.

So, what's the formal definition of independence? Any set of n vectors in $\mathbb{R}^m$ must be linearly dependent if $n > m$.

When $v_1, v_2, \dots, v_n$ are columns of A .

They are independent if nullspace of A is zero vector $\rightarrow \text{rank} = n$

They are dependent if $AC = 0$ for some non-zero C . $\rightarrow \text{rank} < n$

$\hookrightarrow$ There are free variables.

Vectors $v_1, \dots, v_k$ span a space means the space consists of all combinations of those vectors.

$\hookrightarrow$ Basis

$\hookrightarrow$ Basis for a space is a sequence of vectors $v_1, v_2, \dots, v_d$ with 2 properties:

1. They are linearly independent

2. They span the space

Example:

i) Space is $\mathbb{R}^3$

One basis is $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow$ They are independent of course.
They produce the standard basis of $\mathbb{R}^3$.

Another Basis $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 8 \end{bmatrix}$

$\otimes \mathbb{R}^n \rightarrow n$ gives basis if the $n \times n$ matrix with those columns are invertible.

$\otimes$ Given a space, every basis for the space has the same number of vectors.

$\otimes$ Every set of independent vectors can be extended to a basis.

Every spanning set of vectors can be reduced to a basis.

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⊗ All bases for a vector space contain the same number of vectors.

The number of vectors in any basis is the "dimension" of the space.

↳ Dimensions

↳ The number of vectors in any basis.

↳ Dimension is unique for a given subspace, no matter which basis you use.

↳ For $C(A)$, the dimension is equal to the rank (r) of the matrix

↳ Dimension of the column space = Rank of the matrix.

Key points:

① Independent vectors

↳ A set of vectors is linearly independent if none of the vectors can be written as a combination of the others.

↳ No redundancy here - each vector adds new information.

② Spanning a space

↳ A set of vectors spans a space if any vector in that space can be written as a linear combination of the given vectors.

↳ The set provides full coverage of the space

③ Basis for a space

↳ A basis is a set of vectors that is both

↳ A basis is a set of vectors that is both independent and spans the space.

↳ A basis has just the right number of vectors - removing any would make it fail to span, and adding any would create dependence.

Basically, A perfect balance - no redundancy, but complete coverage.

⑨ Dimension of a space

↳ The dimension of a space is the number of vectors in any of its bases.

↳ Basically, the smallest number of vectors needed to describe the space completely.